

The Extension Problem for Orthomodular Observation Algebras

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1 Background

All measurement is finite. An experimenter performs finitely many trials, records finitely many digits, distinguishes finitely many outcomes. Yet the mathematical frameworks we use to organise empirical knowledge — probability theory, quantum mechanics, statistical mechanics — rest on infinite structures: σ -algebras, Hilbert spaces, continuous symmetry groups. The passage from finite observation to infinite structure is not automatic; it requires a commitment that no finite body of evidence can compel.

In probability theory, the locus of this commitment is *countable additivity*. Given a Boolean algebra $\mathcal{E}$ of events and a finitely additive charge $\ell : \mathcal{E} \rightarrow [0, 1]$ with $\ell(\Omega) = 1$, the condition

$$E_n \downarrow \emptyset \implies \ell(E_n) \rightarrow 0 \tag{1}$$

— that mass cannot persist along a vanishing sequence of events — is not derivable from any finite collection of observations. De Finetti [10] argued that only finite additivity is empirically grounded; Kolmogorov [26] adopted σ -additivity as an axiom and built modern probability on it. The question of *when* and *why* the passage from one to the other is justified has never been fully resolved.

In quantum theory, the same tension appears in a different guise. The algebra of propositions about a quantum system is not Boolean: incompatible observables cannot be simultaneously determined. The closed subspaces of a Hilbert space H form an orthomodular lattice $L(H)$ — a structure in which complementation is well-defined but distribution of meet over join may fail:

$$a \wedge (b \vee c) \neq (a \wedge b) \vee (a \wedge c) \quad \text{in general.} \tag{2}$$

A *state* on such a lattice is a function $s : L(H) \rightarrow [0, 1]$ satisfying

$$s(\mathbf{1}) = 1, \quad s(a \vee b) = s(a) + s(b) \quad \text{whenever } a \perp b. \tag{3}$$

The non-distributivity means that s is additive only on orthogonal pairs, not on arbitrary finite unions. The classical machinery for extending finitely additive assignments to σ -additive ones — Carathéodory's outer-measure construction, which requires a Boolean algebra of sets — does not apply.

For the specific lattice $L(H)$, this difficulty is resolved by a remarkable rigidity. Gleason's theorem [19] shows that, when $\dim H \geq 3$, every state has the form

$$s(P) = \text{tr}(\rho P) \tag{4}$$

for a unique density operator ρ , and is therefore automatically σ -additive *on the lattice*. The geometry of Hilbert space is so constraining that, for additivity on $L(H)$ itself, the problem of the infinite does not arise — states have no room to misbehave. (On the *extension* axis —

passage to a measure on the dual $S_0(L(H))$ — even these rigid states fail, by §2.1; the rigidity is descent-side only.)

But Hilbert space is very special. Orthomodular lattices arise in many other settings — the projection lattices of operator algebras, the logics of quantum systems with superselection rules, and the abstract lattices studied in quantum logic — and for these, no analogue of Gleason’s rigidity is known. The question that probability theory and quantum theory share — when does finite coherence force countable behaviour? — remains open in this broader setting, and is the subject of the present note.

The programme “Structure from Observation” [28] approaches this question from a particular direction: rather than beginning with a pre-given space Ω of outcomes and asking which σ -algebras it supports, it begins with the *observations themselves* — a directed system of event algebras $\mathcal{E}_i$ with compatible charges ℓ_i — and asks what the directed structure alone determines.

Paper I answers this for Boolean event algebras. A compatible family of finitely additive charges extends to a σ -additive probability measure if and only if collective exhaustion (CE) holds — condition (1) applied uniformly across the directed system. Two constructions realise the extension:

- **Carathéodory:** assumes CE; produces a measure on $(\Omega, \sigma(\mathcal{C}))$ directly.
- **Stone:** assumes nothing; produces a Baire measure on the Stone space $\text{St}(\mathcal{C})$ unconditionally.

Under discriminability, the two agree. Three questions orient the broader programme:

- (1) What must be added to observation to get structure?
- (2) Reconstruction without points: what structure is already present in the relational description — contexts, directed refinement — before point-spaces are assumed?
- (3) Local-to-global extension: what conditions on the *site* control whether local data extend globally?

Paper I resolves (2) and (3) for Boolean algebras. The present note asks whether any of this survives when the event algebra is orthomodular but not distributive.

Epperson and Zafiris [16] advocate a *relational realist* ontology in which structure is constituted by relations among contexts, not by points in a pre-given space. An orthomodular lattice A is the natural algebraic expression of this: its elements are propositions, its partial order is entailment, and its non-distributivity reflects the impossibility of assigning simultaneous truth values to all propositions. The dual space $S_0(A)$ of McDonald–Bimbó [29] is built from *filters* — consistent collections of propositions — not from points of a pre-existing space. This is question (2) applied to non-Boolean algebras: what structure does the relational description already contain?

In the Boolean case, one might hope that the passage from finite to countable additivity could be understood as a *sheaf condition* — local states gluing to a global σ -additive measure. Biesel [1] shows this hope is disappointed:

- Finitely additive charges form a sheaf for finite covers.
- σ -additive measures form a sheaf for countable covers.
- The step between is the *definition* of σ -additivity, not a structural theorem.

This is the boundary at which the present problem begins.

Paper II [27] introduces three positions on the relationship between observation and reality, each a constraint on where the dual-space measure lives:

Empirical adequacy (EA)

The dual-space measure agrees with all observations.

Probabilistic realism (PR)

The measure descends to a realisation space.

Value-definite realism (VDR)

A global two-valued homomorphism exists.

These are nested: $VDR \Rightarrow PR \Rightarrow EA$. For Boolean algebras, all three are commensurable — every EA-theory admits a VDR-completion. For OMLs with $\dim \geq 3$, Gleason’s theorem (4) secures a σ -additive measure on the *lattice* $L(H)$, while Kochen–Specker [25] blocks VDR. The extension problem sits at the $EA \rightarrow PR$ transition: can we pass from a state on the algebra (EA) to a σ -additive measure on the dual space (PR)? Paper II defines PR as a measure on an underlying state space obtained by *descent from the dual-space measure*, so PR presupposes the dual-space measure exists. §2.1 shows that for $L(H)$ this presupposition *fails*: no normal state yields a charge on $S_0(L(H))$, so the dual-space measure does not exist and PR cannot be reached — the chain breaks at the $EA \rightarrow PR$ extension step, not at descent. This **contradicts Paper II’s Commensurability theorem (b)**, which attributes the $L(H)$ dual-space measure to Gleason; §2.1 shows Gleason gives only the *lattice* measure, not the dual-space one. The $L(H)$ verdict flips from “PR holds (Gleason)” to “PR fails for normal states” — a correction to Paper II pending the author’s decision (see the session record), not resolved here. Under what conditions, then — and is descent a free choice (as in the Boolean case) or a structural consequence of the algebra and state together?

Van Fraassen’s distinction [37] between empirical adequacy and structural commitment, which Paper I makes precise via the Stone construction, thus acquires a sharper form in the OML setting: the Stone route is no longer available, the Carathéodory route is blocked by non-distributivity, and the passage from coherent local data to global probability — question (3) — requires genuinely new ideas.

2 The problem

Definitions. We collect the lattice-theoretic notions used throughout. All are standard except *valuation*, where we adopt a local convention flagged below; the elementary laws are exactly those verified for MO_3 in `verification/mo3_extension.py`.

Definition 2.1 (Orthocomplemented lattice). A *bounded lattice* is a partially ordered set in which every pair a, b has a least upper bound $a \vee b$ (join) and greatest lower bound $a \wedge b$ (meet), with a greatest element $\mathbf{1}$ and least element $\mathbf{0}$. An *orthocomplementation* is a map $a \mapsto a^\perp$ satisfying, for all a, b :

- (i) $a \wedge a^\perp = \mathbf{0}$ and $a \vee a^\perp = \mathbf{1}$ (complement),
- (ii) $a^{\perp\perp} = a$ (involution),
- (iii) $a \leq b \Rightarrow b^\perp \leq a^\perp$ (order-reversing).

Elements a, b are *orthogonal*, written $a \perp b$, if $a \leq b^\perp$ (equivalently $b \leq a^\perp$). Orthogonality is strictly stronger than *meet-zero* ($a \wedge b = \mathbf{0}$) once the lattice is non-distributive — the gap that drives this note.

Definition 2.2 (Orthomodular lattice). An orthocomplemented lattice A is *orthomodular* if it satisfies the *orthomodular law*:

$$a \leq b \quad \Longrightarrow \quad b = a \vee (a^\perp \wedge b). \quad (5)$$

This is the modular law restricted to the case where the smaller element is the orthocomplement: it says a together with “the part of b orthogonal to a ” recovers b . A *Boolean algebra* is the special case where $\wedge$ distributes over $\vee$; every Boolean algebra is orthomodular, but not conversely. The motivating non-Boolean example is $L(H)$, the lattice of closed subspaces of a Hilbert space H with $a^\perp$ the orthogonal complement; the smallest is MO_3 (Figure 1).

A *state* on A is a map $s : A \rightarrow [0, 1]$ with $s(\mathbf{1}) = 1$ that is additive on orthogonal pairs, $s(a \vee b) = s(a) + s(b)$ whenever $a \perp b$ — exactly (3). Three strengthenings are distinguished in §2 below; one needs a word of warning.

Definition 2.3 (Valuation, *local convention*). We call a state s a *valuation* if it is additive on every *meet-zero* pair: $s(a \vee b) = s(a) + s(b)$ whenever $a \wedge b = \mathbf{0}$ (not merely $a \perp b$). **Warning:** this is *not* the standard lattice-theoretic “valuation” ($v(a) + v(b) = v(a \wedge b) + v(a \vee b)$); we use the term in this local sense throughout, for the condition sitting strictly between *state* and *extends-to-a-charge*.

Definition 2.4 (OMP; concrete logic). An *orthomodular poset* (OMP) weakens Definition 2.2: joins $a \vee b$ are required to exist only for *orthogonal* pairs (orthogonality and orthomodularity are then imposed where the joins exist). An OMP is a *concrete logic* (equivalently *set-representable*) if it embeds into the orthocomplemented poset $(\mathcal{P}(X), \subseteq, X \setminus (-))$ of subsets of some set X , with $\mathbf{0} = \emptyset$, $\mathbf{1} = X$, orthocomplement the set complement, and the embedding preserving existing orthogonal joins.

Let A be an orthomodular lattice (OML) and let $s : A \rightarrow [0, 1]$ be a state (Definition 2.3 and (3)). The McDonald–Bimbó duality [29] associates to A a compact topological space

$$S_0(A) = (F(A), \subseteq, \perp_A, P(A), T(S)),$$

where $F(A)$ is the set of filters on A and $P(A) \subseteq F(A)$ is the set of prime filters. For each $a \in A$, set

$$h(a) = \{x \in F(A) : a \in x\}.$$

The map h is an OML isomorphism from A onto the $\perp$ -stable clopen sets $\text{CO}(S_0(A))^\dagger$. Define a set function μ on these clopens by

$$\mu(h(a)) = s(a). \tag{6}$$

Question 2.5 (Extension). Under what conditions on the OML A does μ extend to a σ -additive measure on the Baire (or Borel) σ -algebra of $S_0(A)$?

This is the direct OML analogue of the Stone extension theorem established for Boolean algebras in [28].

In the Boolean case, the extension is automatic. The clopens $\text{Clop}(\text{St}(\mathcal{C}))$ form a Boolean algebra, the state s restricts to a finitely additive charge on this algebra, and the Carathéodory extension theorem (or Choksi’s projective-limit argument [8]) produces the σ -additive extension. Compactness of the Stone space gives σ -subadditivity for free.

In the OML case each of these steps breaks:

- (i) $\text{CO}(S_0(A))^\dagger$ is an OML, *not* a Boolean algebra.
- (ii) The state s is additive on orthogonal pairs only, not on arbitrary finite unions.
- (iii) **Carathéodory does not apply** — there is no Boolean algebra of sets on which to run the outer-measure construction.
- (iv) Compactness holds (McDonald–Bimbó, Lemma 3.5 [29]), but the σ -subadditivity argument requires orthogonality structure that the topology alone does not provide.

The gap is sharp: “finitely additive on a compact Boolean algebra of clopens” automatically extends; “orthogonally additive on a compact OML of clopen $\perp$ -stable sets” does *not*.

For the prototypical case $A = L(H)$ with $\dim H \geq 3$, Gleason’s theorem (4) gives a σ -additive measure on the *lattice* $L(H)$ itself: every state is induced by a density operator. But

this lattice-level rigidity does *not* transfer to the *extension* axis — the passage to a charge on the dual $S_0(L(H))$ — which §2.1 shows fails for every normal state. (The two-axis split below is exactly what separates “measure on $L(H)$ ” from “measure on $S_0(L(H))$.”) And since extension fails, no dual measure exists to descend, so the descent question does not arise for normal states on $L(H)$. For general OMLs, no Gleason-type representation is available even at the lattice level. Question 2.5 asks for the structural conditions that substitute for the Hilbert-space geometry.

Feature	Boolean (Paper I)	OML (this problem)
Dual space	$\text{St}(A)$ (ultrafilters)	$S_0(A)$ (all filters)
Distinguished subset	$\text{pure}(\Omega)$, not canonical	$P(A)$, canonical
Clopens	Boolean algebra	OML (non-distributive)
Additivity of s	full (finite)	orthogonal pairs only
Extension (to the dual)	automatic (Carathéodory)	fails for normal states (§2.1); Gleason is descent-side
Realisation constrained?	No	Yes (Kochen–Specker [25])

Table 1: Structural comparison of the extension problem.

Two axes: extension and descent. The passage from a state on A to a measure on $S_0(A)$ splits into two axes that the Boolean case fuses but the OML case separates. Since $S_0(A)$ is a Stone space, its *full* clopen algebra $\text{CO}(S_0(A))$ is Boolean, whereas the state lives only on the $\perp$ -stable clopens $\text{CO}(S_0(A))^\dagger$ — the OML, the image of h . These share only meet; join, complement, and bottom differ.

Extension Extend μ from the $\perp$ -stable clopens to a finitely additive charge on the *full* Boolean algebra of clopens — the classical Boolean extension problem [23]. Since h preserves meets, $a \wedge b = 0$ forces $h(a) \cap h(b) = h(0)$, so meet-zero elements of A map to *disjoint* clopens. A charge must therefore satisfy, for every pairwise-meet-zero family $\{a_i\}$, $\sum_i s(a_i) \leq 1$ — a Pitowsky correlation polytope / Bell–Boole inequality system [31]. σ -additivity of the state does not help here.

Compactness

Once extension succeeds, the charge extends to a σ -additive Borel measure automatically. Unconditional.

In the Boolean case (Paper I), the extension axis is *free*: finite additivity on the clopen Boolean algebra plus compactness gives the Stone measure unconditionally — “the first arrow is free.” In the OML case, **the extension axis is no longer free.**

The extension obstruction is meet-zero versus orthogonal. Three strengthening conditions on s must be distinguished (Pták–Pulmannová [34], Def. 2; cf. Definitions 2.1–2.3):

- (1) *State*: additive on *orthogonal* pairs ($a \leq b^\perp$).
- (2) *Valuation* (local sense, Def. 2.3): additive on *meet-zero* pairs ($a \wedge b = 0$) — a binary condition, strictly stronger, since in a non-distributive OML meet-zero is coarser than orthogonality.
- (3) *Extends to a Boolean charge*: the n -ary condition $\sum_i s(a_i) \leq 1$ for every pairwise-meet-zero family — strictly stronger again.

The extension axis is condition (3). It can fail for *every* state. On the smallest non-distributive OML MO_3 (three 2×2 blocks pasted at 0 and 1), all six atoms are pairwise meet-zero, so orthoadditivity forces the three complementary pairs to sum to 3 while a charge demands ≤ 1 : no state extends. The maximally mixed state $s \equiv \frac{1}{2}$ is a valuation (it satisfies the binary condition (2)) yet already fails the n -ary condition (3) at the triple $\{p, q, r\}$, where $\frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} > 1$. So the obstruction is the gap between meet-zero and orthogonality, not σ -additivity and not non-contextuality.

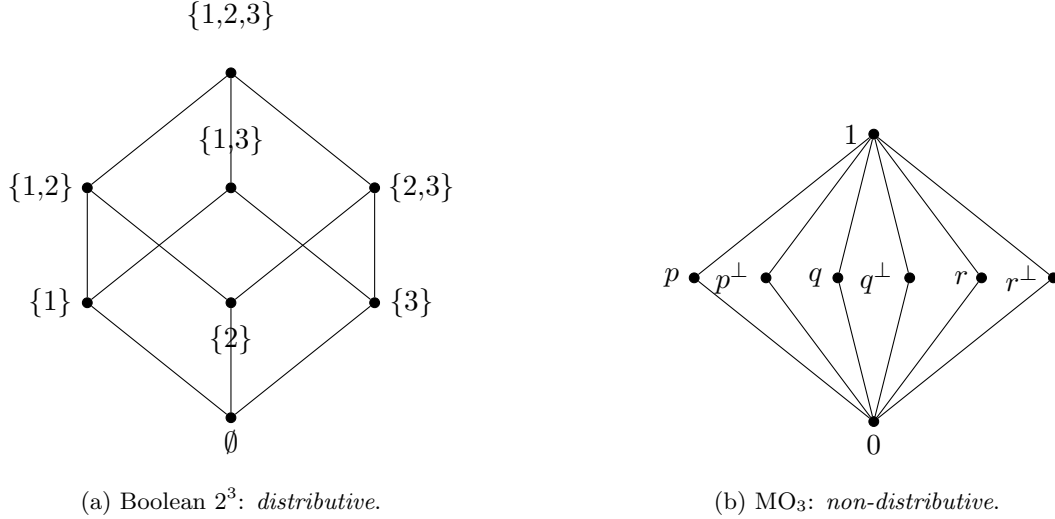


Figure 1: Hasse diagrams (edges are covering relations). In the Boolean cube (a), each singleton lies below the *two* pairs that contain it; those four-cycles are exactly distributivity, and complementation $\{i\} \mapsto \{j, k\}$ matches the geometry. In MO_3 (b) every non-trivial element is *both* an atom and a coatom: the lattice has height 3, the three complementary pairs $(p, p^\perp), (q, q^\perp), (r, r^\perp)$ share only 0 and 1, and any two distinct atoms have meet 0 and join 1. Distributivity fails at once: $p \wedge (q \vee r) = p \wedge 1 = p$, but $(p \wedge q) \vee (p \wedge r) = 0 \vee 0 = 0$. The six atoms being pairwise meet-zero is what forces $s(p) + s(q) + s(r) \leq 1$ on any extending charge, which $s \equiv \frac{1}{2}$ violates. Structure verified in `verification/mo3_extension.py`.

The concrete/non-concrete boundary is not where the gap closes. A set-representable OMP (P, L) requires only closure under complement and under *disjoint* union (Burešová–Pták [4], Def. 1.1); closure under intersection is *not* required — it would force L to be a field of sets, hence Boolean. The lattice meet $a \wedge b$ is the greatest L -member contained in $A \cap B$, so $a \wedge b \subseteq A \cap B$ with equality iff $A \cap B \in L$. Hence orthogonality ($A \cap B = \emptyset$) implies meet-zero always, but **meet-zero does not imply orthogonality** for incompatible pairs. The smallest witness is MO_2 , the concrete logic on $P = \{1, 2, 3, 4\}$ with $L = \{\emptyset, \{1, 2\}, \{3, 4\}, \{1, 3\}, \{2, 4\}, P\}$: there $\{1, 2\} \wedge \{1, 3\} = 0$ while $\{1, 2\} \cap \{1, 3\} = \{1\} \neq \emptyset$. Crucially, MO_3 *itself is concrete* (it carries 2^3 ordering two-valued states, hence is set-representable by Gudder’s representation theorem for concrete logics — a concrete logic is exactly an OMP with an order-determining set of two-valued states [33]), so the flagship “meet-zero $\neq$ orthogonal” example above is a concrete logic. The boundary that matters is therefore not concrete versus non-concrete but a *richness* / atomicity condition: meet-zero coincides with orthogonality iff every nonempty $A \cap B$ contains a nonzero member of L (sufficient: all singletons lie in L). This property has no established standard name (its cousins — Jauch–Piron, Tkadlec “regional,” Burešová–Pták “point-distinguishing” — are all distinct). (Terminology trap: MO_3 ’s celebrated non-representability is von Neumann *coordinatisation* — failure to be a subspace lattice of a projective geometry — a different notion from set-representability.)

Status of the extension axis. For *rich* concrete OMLs (singletons in L , or the weaker intersection-richness above) meet-zero coincides with orthogonality, and the extension axis reduces to the classical marginal / Pitowsky non-contextuality problem. For *non-rich* concrete OMLs (MO_2 , MO_3) and non-concrete OMLs ($L(H)$) the meet-zero/orthogonal gap is non-empty, and the extension axis is strictly stronger and can fail universally. In the *finite* case it is a decidable linear program, not a structurally resistant frontier. The *infinite* case — $L(H)$, to which the finite counterexample does not directly speak — is resolved for normal states by the clustering argument of §2.1 below, with only the singular states left open.

2.1 The infinite extension case for $L(H)$

For $L(H)$ with $\dim H = \infty$ the extension axis admits a decisive finite-dimensional argument, sharper than the MO_3 linear program. Fix a 2-dimensional subspace $e \subset H$. For any line $p \subset e$, its perpendicular $p^\perp \subset e$ satisfies $p \perp p^\perp$ and $p \vee p^\perp = e$, so orthoadditivity gives

$$s(p) + s(p^\perp) = s(e). \quad (7)$$

Choose k distinct lines $p_1, \dots, p_k \subset e$ in general position, so that the $2k$ lines $\{p_i, p_i^\perp\}$ are pairwise distinct. Distinct lines meet only at 0, hence are pairwise meet-zero, so h sends them to $2k$ pairwise-disjoint clopens. A finitely additive charge extending $\mu(h(a)) = s(a)$ must therefore satisfy, using (7),

$$\sum_{i=1}^k [s(p_i) + s(p_i^\perp)] = k s(e) \leq 1 \quad \text{for every } k \geq 1. \quad (8)$$

If $s(e) > 0$, taking $k > 1/s(e)$ violates (8). Hence:

Proposition 2.6. *Any state s on $L(H)$ with $s(e) > 0$ for even one finite-dimensional e fails to extend to a charge on the full clopen algebra of $S_0(L(H))$. In particular no normal (Gleason) state extends. The argument uses only orthoadditivity on lines in a single 2-plane; σ -completeness of the lattice is irrelevant.*

Every normal state $s = \text{tr}(\rho \cdot)$ has $s(e) > 0$ for some 2-dim e (as $\rho \neq 0$), so Proposition 2.6 kills the entire σ -additive (Gleason) state space. This answers the infinite extension question for the physically central states, and answers “does σ -completeness change it?” in the negative.

The only states the argument does *not* reach are the *singular* ones: $s(e) = 0$ for every finite-dimensional e , where the bound (8) is vacuous. These exist (take H separable) — e.g. the ultrafilter vector state $s(a) = \lim_\omega \langle a e_n, e_n \rangle$ for a free ultrafilter ω and an orthonormal basis (e_n) , or any pullback of a state on the Calkin algebra $B(H)/K(H)$. Moreover, lifting a lattice state to a bounded functional φ on $B(H)$ by Mackey–Gleason / Bunce–Wright [3, 9] (valid as $B(H)$ has no type I_2 summand), the Takesaki normal/singular decomposition $\varphi = \varphi_n + \varphi_s$ with both parts positive [36] writes $B(H)^* = (\text{trace class}) \oplus K(H)^\perp$ — where the singular part φ_s annihilates the compacts $K(H)$ (a statement at the level of φ , after the lift, not of the bare lattice state). A finite-rank projection e then has $s(e) = \text{tr}(\rho e)$, and $s(e) = 0$ for all finite-dim e forces $\rho = 0$ (taking e rank-one gives $\langle \rho \xi, \xi \rangle = 0$ for all ξ). So a state is *either* detected by some finite-rank projection (killed by Proposition 2.6) *or* purely singular — a clean dichotomy with no third class (the decomposition being unique). The density $\varphi_n = \text{tr}(\rho \cdot)$ is named by predual duality $B(H)_* = \text{trace class}$, *not* by Gleason. Whether a singular orthoadditive state extends is the open residue; there the relevant meet-zero families are infinite-dimensional subspaces with trivial intersection, and σ -additivity *of the state* (not of the lattice) is what bites.

The hinge — and the first thing to settle before deciding whether this residue is genuinely more tractable than the descent residue (§4) — is whether the singular obstruction is reached with *finite* or *countable* meet-zero families. The normal-state argument (Proposition 2.6) is

finitary: k lines, k arbitrary but finite, and the bound is violated at finite k . This is why it needs no σ -duality and dodges the statability wall of §4. But the phrase “ σ -additivity of the state is what bites” suggests the singular obstruction may instead require a *countable* family of infinite-dimensional subspaces — in which case it is *not* cleanly finitary and does not obviously dodge that wall. If the obstruction is finite, the singular case is a self-contained operator-algebra question (Calkin / Bunce–Wright machinery) settleable ahead of the descent residue; if countable, it is entangled with the same finitary-to- σ bridge and is not clearly the easier of the two. Settling *which* is the concrete next step; it is mathematical work, not yet done here. (Proposition 2.6 is the proof; the inequality is instance-checked — finitely many lines in $\mathbb{R}^2$, the charge LP infeasible once $k s(e) > 1$ — by `verification/lh.infinite_extension.py`, which does not itself touch the infinite-dimensional $S_0(L(H))$). The singular/normal dichotomy is hand-verified — foundations cited with hypotheses checked, gluing checked by hand — in `verification/lh.singular_dichotomy.md`.)

Descent. σ -additivity of the state governs descent, not extension. Even once μ extends to a measure $\hat{\mu}$ on $S_0(A)$, the *descent* question — whether $\hat{\mu}$ concentrates on the “physical” points — changes character:

Boolean	Does $\hat{\mu}$ concentrate on $\text{pure}(\Omega)$? If and only if the charges are σ -additive. The choice of realisation space Ω is unconstrained.
OML	Does $\hat{\mu}$ concentrate on $P(A)$? Here $P(A)$ is part of the structure, not a free choice. The Kochen–Specker obstruction [25] prevents simultaneous realisation of all filters; the Bub–Clifton theorem [2] shows that the state determines a maximal Boolean subalgebra of definite values.

In the OML setting, the algebra and state *together* constrain which filters are “realised.” Descent is not a geometric commitment but a structural consequence.

Remark 2.7 (Decomposition lives on the descent axis). De Simone–Navara [11] decompose OMP states as $s = s_\sigma + s_{\text{wpfa}}$ (a σ -additive part and a weakly purely finitely additive part), the orthomodular analogue of Yosida–Hewitt. By the axis split above, the wpfa component governs the *descent* axis — it is the analogue of the purely finitely additive part ℓ_p in the Boolean case — and *not* the extension axis. A natural-looking conjecture, “ s extends iff $s_{\text{wpfa}} = 0$,” is therefore mis-targeted: vanishing of s_{wpfa} concerns whether the (already σ -additive) state’s measure concentrates on $P(A)$, while extension is blocked by non-distributivity regardless of the wpfa component. The decomposition theory and the extension problem live on different axes.

A caveat compounds this. The McDonald–Bimbó duality [29] is *finitary*: the identity $h(\bigvee_n a_n) = (\bigcup_n h(a_n))^{\perp\perp}$ holds for finite joins but fails in general for countable joins, since finitary OML homomorphisms need not preserve infinite suprema. Making even the descent argument rigorous — the analogue of Paper I’s “ σ -additivity $\Leftrightarrow$ concentration” — requires σ -completeness and continuity hypotheses the 2023 duality does not supply, and is itself open.

The finitary-to- σ bridge: open, with three named routes blocked. Paper I’s descent runs on a *Loomis–Sikorski* engine: σ -additive measures on the Stone space correspond to σ -homomorphisms off the ideal points, which is what makes “ σ -additive $\Leftrightarrow$ concentrates on $\text{pure}(\Omega)$ ” go through. The OML descent question is whether that engine has an OML analogue — whether a σ -complete OML admits a Loomis–Sikorski representation (quotient of a σ -tribe of sets by a σ -ideal) or a countable-join-preserving σ -Stone duality. The literature settles only the known *routes*, all of which are blocked.

- (i) **RDP / effect-algebra route — blocked.** Loomis–Sikorski holds for σ -complete MV-algebras (Dvurečenskij [14]) and for monotone σ -complete effect algebras *with* the Riesz

Decomposition Property (RDP) [15]. But a lattice effect algebra has RDP iff it is an MV-algebra (Riečanová; Paseka), and an OML that is an MV-algebra — all pairs compatible — is Boolean (Kalmbach [24]). So $OML + RDP \Leftrightarrow \text{Boolean}$: the RDP machinery has no non-Boolean OML to act on.

- (ii) **MacNeille-completion route — blocked.** Harding [20] shows the MacNeille completion of an OML need not be orthomodular, so one cannot complete to absorb countable joins.
- (iii) **σ -Stone-duality route — none exists.** McDonald–Bimbó is irreducibly finitary; Freytes’ equational theory of σ -complete OMLs [18] supplies no set/tribe representation.

A concentration statement needs a point space, which a general OML may lack (Greechie stateless lattices are non-concrete). But that is a red herring for the live case: $L(H)$ is non-concrete yet Gleason hands it a clean lattice measure (the non-concreteness is no obstacle there), and MO_3 is concrete. The genuinely open class is **concrete, non-Boolean, infinite σ -OMLs**, where the non-concreteness obstruction is absent and the RDP argument blocks only one route. No impossibility theorem is known for this class; the contribution here is the map of dead routes, which sharpens “needs a new idea” into “needs a representation bypassing all three.”

What is at stake: relational probability without realizations. The descent residue carries a motivation beyond technical completeness. Paper I’s finding was that *realization* — which points are “real” — is structure added on top of a coherent assignment, not forced by it. In the McDonald–Bimbó setting, realization is concentration on the dual points $P(A) \subseteq S_0(A)$. A construction of σ -additive probability on a non-distributive OML *that does not pass through such a concentration* would be probability built from the entailment relation alone — a relational, point-free probability, the non-Boolean analogue of the localic (pointless) measures distributivity already permits.

Two cautions keep this honest. First, the separation already established for $L(H)$ — a σ -additive measure exists on the lattice (PR_{lat} , by Gleason) while no measure concentrates on the dual points (PR_{dual} fails, §2.1) — shows a lattice measure *need not descend to the McDonald–Bimbó points*. But it does *not* exhibit point-free probability: “point” is overloaded. Gleason removes the *dual* points $P(A)$ while resting entirely on the *Hilbert* points (the rays of H , from which the density ρ in $s = \text{tr}(\rho \cdot)$ is built); indeed Gleason is a *representation* theorem, reducing every lattice state to that point datum. So $L(H)$ witnesses the separation, not point-freeness, and cannot serve as the existence proof we want. Second, Gleason fires only because H is present; for a general OML there is no such theorem. Hence a genuinely point-free construction must be built, not borrowed — and the only route to it is exactly the missing σ -OML representation of the route-map above. This is the substantive reason to chase residue #2: it is the engine for relational probability without realizations, and — since $L(H)/\text{Gleason}$ does not supply that witness — the *only* candidate engine, not a redundant one.

Two grades of probabilistic realism. The two axes, re-expressed in the EA/PR/VDR vocabulary of Paper II [27], split *probabilistic realism* (PR) into two grades that the Boolean case fuses and non-distributivity separates — the same extension/descent split, named in the realism language, not a new construction.

- | | |
|--------------------------------------|--|
| PR_{lat} | A σ -additive measure on the <i>lattice</i> A agreeing with s . For $L(H)$, $\dim H \geq 3$, Gleason supplies this for every σ -additive state. |
| PR_{dual} | A σ -additive Borel measure $\hat{\mu}$ on the <i>dual</i> $S_0(A)$ with $\hat{\mu}(h(a)) = s(a)$ concentrating on the physical points, $\hat{\mu}(P(A)) = 1$ — extension (§2) followed by descent. |

The relation between the two grades is a *witnessed one-directional separation*, not a proven nesting. $L(H)$ normal states *separate* them: they clear PR_{lat} (Gleason) and fail PR_{dual} (§2.1:

no charge on $S_0(L(H))$ exists, a fortiori none concentrating). The reverse inclusion $\text{PR}_{\text{dual}} \Rightarrow \text{PR}_{\text{lat}}$ is *not* free: restricting a dual measure $\hat{\mu}$ to the $\perp$ -stable clopens recovers s only as a *finitely* orthoadditive state, whereas PR_{lat} needs σ -additivity on the lattice, $s(\bigvee_n a_n) = \sum_n s(a_n)$. The gap is $\hat{\mu}(D)$, $D = h(\bigvee_n a_n) \setminus \bigcup_n h(a_n)$, and concentration on $P(A)$ does *not* force $\hat{\mu}(D) = 0$: D contains *principal* filters (e.g. $\uparrow p$ with $p = \text{span}(v)$, v of infinite support, $a_n = \text{span}(e_n)$: $p \leq \bigvee_n a_n$ but $p \not\leq a_n$ for all n). So $\text{PR}_{\text{dual}} \Rightarrow \text{PR}_{\text{lat}}$ requires an extra σ -continuity hypothesis ($\hat{\mu}$ null on countable-join defects) that the finitary McDonald–Bimbó duality does not supply — the *same* finitary wall flagged above (verified against the primary source, `verification/mb_primeness_check.md`). What *is* established is the separation, and it lets one assert *both* grades at once — Gleason clears PR_{lat} for $L(H)$ while PR_{dual} fails — without suppressing either. In the Boolean case the two grades coincide (Stone gives $\hat{\mu}$ unconditionally; descent $\Leftrightarrow \sigma$ -additivity), which is why Paper II could write a single PR. The distinction is philosophically neutral: a Born-rule realist reads off PR_{lat} , a relational realist PR_{dual} , a constructive empiricist EA.

What is *not* established is that descent does any work independent of extension. Every known PR_{dual} failure is an extension failure; in finite dimensions descent is vacuous (the only non-principal point is the improper filter, already null by $s(0) = 0$ — verified, `verification/pr_dual_inhabita` where PR_{dual} on MO_2 collapses to the extension LP). The descent half can be strictly weaker than extension only in infinite dimensions, where free non-principal filters can carry mass — exactly where the finitary-to- σ bridge above fails. Hence the open question: *is there a σ -OML regime in which descent is both statable and strictly weaker than extension — a non-distributive OML and state that extends to a charge on $S_0(A)$ but fails to concentrate on $P(A)$?* The only candidate is the singular-state sliver of §2.1, which runs into the same statability wall. Until such a witness exists, $\text{PR}_{\text{lat}}/\text{PR}_{\text{dual}}$ is a witnessed one-directional *separation* ($L(H)$ normal states clear PR_{lat} , fail PR_{dual}) — *not* a proven nesting, since $\text{PR}_{\text{dual}} \Rightarrow \text{PR}_{\text{lat}}$ needs the same σ -continuity hypothesis the finitary duality withholds — with the descent half not yet shown independent: a vocabulary refinement, not a new theorem.

3 What is known

In the Boolean case, the Kelley–Vladimirov–Pták characterisation (Fremlin, Theorem 391D [17]) gives a complete algebraic answer:

Theorem 3.1 (KVP). *A Boolean algebra B carries a strictly positive σ -additive measure if and only if B is*

- (a) *Dedekind σ -complete,*
- (b) *weakly (σ, ∞) -distributive, and*
- (c) *chargeable (carries a strictly positive finitely additive measure).*

Each condition is algebraic and non-trivially constraining. For OMLs, the analogues fail or degenerate:

- (a) Dedekind σ -completeness makes sense but is very strong.
- (b) Weak (σ, ∞) -distributivity has no clean OML analogue — the definition relies on distributivity of $\bigwedge$ over $\bigvee$, which may fail orthomodularly.
- (c) Chargeability has no known structural characterisation on OMLs; the existence of a strictly positive finitely additive measure cannot be read off from the lattice structure.

The positive results that do exist for OMLs all require algebraic structure rich enough to approximate Hilbert-space geometry:

Gleason [19] $A = L(H)$, $\dim H \geq 3$. Every state has the form (4); σ -additivity on the lattice is automatic (this is lattice-level, not dual-space extension — cf. §2.1).

Bunce–Wright [3] *A* the projection lattice of a JBW-algebra. σ -additivity via operator-algebraic structure.

Chetcuti–Dvurečenskij [7]
Lattice effect algebras with completeness conditions close to von Neumann algebras.

The obstructive results show that the Boolean extension machinery cannot be transplanted:

Pták–Pulmannová [33]
If every unital subadditive measure on an orthomodular poset is a state, the poset must be Boolean. This is the structural reason no KVP analogue exists for OMLs: conditions strong enough to force σ -additivity collapse the lattice back to a Boolean algebra.

Navara [30] Regularity does *not* force σ -additivity on general orthomodular posets. The Béaver–Cook result is special to von Neumann algebras.

Pták [32] Exotic OML state spaces exist; the obstruction is combinatorial (Greechie pasting), not from the centre.

Recent incremental work confirms the field is active but no one has found the right condition: Voráček–Pták [38] study signed measures on orthomodular posets; Burešová–Pták [5] investigate variations on regularity conditions; De Simone–Navara study Yosida–Hewitt-type decompositions for orthomodular posets.

For the σ -additivity / descent side, the gap between “too weak” (allows non- σ -additive states) and “too strong” (collapses to Boolean) appears structurally robust — this is the part where further reading has not produced the answer. (This robustness is about the descent axis; the extension axis is classical, per §2.)

Prior art on point-free / contextual quantum probability, and where the gap sits.
The point-free and directed-context constructions of quantum probability in the literature each land on one side of a single line — the σ -additivity-on-a-non-distributive-structure line — and none crosses it.

Gleason tradition

Varadarajan, Kalmbach, Pták–Pulmannová: σ -additive measures on a genuinely non-distributive OML, but real-valued on a *fixed* lattice — not point-free and not built from a directed observation system.

Döring 2009 [12] builds measures point-free on the spectral presheaf, indexed over the directed poset of abelian subalgebras — but they are provably only *finitely additive* (a consequence of linearity of the underlying state), and live on the distributive Heyting algebra of clopen subobjects.

Heunen–Landsman–Spitters

[21, 22] (Bohrification) build point-free quantum probability over the directed poset of commutative subalgebras, and their internal valuation is σ -additive — it is a *continuous* (Scott-continuous) valuation, $\mu(\bigvee_i U_i) = \bigvee_i \mu(U_i)$ on directed families. But it factors through an internally *distributive* locale: externally (their Def. for Rickart C^* -algebras) the measure is σ -additive only *on each countably complete Boolean sublattice* of $\text{Proj}(A)$, glued over the context poset. Non-distributivity is tamed by retreat to commutative contexts, not carried natively.

Localic valuations

Vickers, Simpson, Coquand–Spitters: point-free σ -additive (Scott-continuous) probability, but valuations are defined on *frames*, distributive by definition; no orthomodular analogue exists in this tradition.

OML Stone dualities

Two finitary dualities for OMLs exist: McDonald–Bimbó [29] (filter spectrum, used here) and the spectral-presheaf duality of Cannon and Döring [6]. Both are *purely structural* — neither carries a measure, state, or valuation on the non-distributive dual, and neither has a countable-join-respecting (σ -) version.

The descent residue is exactly the unoccupied conjunction: a *point-free*, σ -*additive* measure on a *natively non-distributive* OML, built from the directed observation system, with realization (concentration on $P(A)$) as separable structure. Döring gives up σ -additivity; Bohrification and the localic tradition give up native non-distributivity (retreating to distributive contexts/locales); the Gleason tradition gives up point-freeness. The McDonald–Bimbó filter route is chosen over Cannon–Döring precisely because it carries the principal filters $P(A)$ as explicit structure — the realization datum the programme treats as separable. The contribution, were it to clear its bar, is therefore the σ -OML representation itself together with the collective-exhaustion extension mechanism delivering σ -additivity point-free where Döring’s reaches only finite additivity; the *directed-context-yields-non-distributivity* idea on its own is *not* new (it is the shape of both Bohrification and, purely structurally, the colimit-over-Boolean-contexts generation of OMLs).

All known exotic OML state spaces are built by **Greechie pasting** [39, 35, 32]. Progress on the descent residue likely requires either:

- (a) a new pasting technique that controls σ -additivity, or
- (b) an approach that bypasses the state space entirely (e.g., categorical or topos-theoretic [13]).

4 Assessment

Question 2.5, once split into its two axes, is not a single open problem but a settled axis and an open one.

Extension axis: largely settled, and partly degenerate. This is the classical Boolean extension problem [23] / Pitowsky polytope feasibility [31]. Because h preserves meets, it is governed by the gap between meet-zero and orthogonality, which is non-empty whenever the concrete representation is non-rich (and a fortiori in any non-distributive OML). In the finite case it is a decidable linear program; it can fail for *every* state (MO_3). This is not a structurally resistant frontier and does not require a new idea. Two earlier framings of this axis were mistaken. (i) “The Pták–Pulmannová frontier needing a new idea” — Pták–Pulmannová [34] concerns the *supply* of valuations (a unital set forces Boolean), not whether a given state extends, and extension is in any case strictly stronger than the valuation condition. (ii) “Concrete OMLs reduce the axis to Pitowsky non-contextuality because meet = intersection” — false: concreteness requires only disjoint-union closure, and MO_2 , MO_3 are concrete logics in which meet-zero $\neq$ orthogonality (§2). The gap closes only under a richness / atomicity hypothesis, not under concreteness.

Descent axis: genuinely open, dead routes mapped. Whether a measure on $S_0(A)$ concentrates on the physical points $P(A)$, and the analogue of Paper I’s “ σ -additivity $\Leftrightarrow$ concentration,” cannot even be stated rigorously without σ -completeness and continuity hypotheses, because the McDonald–Bimbó duality is finitary (the countable-join identity fails). The descent argument runs on a Loomis–Sikorski engine, and the three known routes to an OML analogue

are all blocked — the RDP / effect-algebra route ($\text{OML} + \text{RDP} \Leftrightarrow \text{Boolean}$), the MacNeille completion (Harding), and a σ -Stone duality (none exists; Freytes gives only an equational theory). But *no impossibility theorem* is known for the live class — concrete, non-Boolean, infinite σ -OMLs. This is the live residue, now sharpened from “needs a new idea” to “needs a representation bypassing all three blocked routes.”

The infinite extension case: no normal state extends; the singular case is the open sliver. For $L(H)$ with $\dim H = \infty$ the extension fails decisively for every normal state, by a clustering argument inside a single 2-plane (§2.1). Fix a 2-dim subspace e ; for k distinct lines $p_1, \dots, p_k \subset e$, the $2k$ lines $\{p_i, p_i^\perp\}$ are pairwise meet-zero, so a charge forces $\sum_i [s(p_i) + s(p_i^\perp)] = k s(e) \leq 1$ for every k . Hence any state with $s(e) > 0$ for even one finite-dim e fails — which is every normal (Gleason) state. σ -completeness of the lattice is irrelevant; extension is a finitary charge condition. The only escapees are *singular* states ($s(e) = 0$ for all finite-dim e); these exist (e.g. ultrafilter vector states, or pullbacks of Calkin-algebra states) and form, with the normal states, a clean dichotomy (Takesaki normal/singular decomposition via the Bunce–Wright lift [3]). Whether a singular orthoadditive state extends — where the relevant meet-zero families are infinite-dim subspaces with trivial intersection, and σ -additivity of the state is what bites — is the open sliver.

Novelty. The formulation via McDonald–Bimbó duality [29] appears to be new; the identification of the extension axis with classical Horn–Tarski / Pitowsky feasibility de-mystifies it. For $L(H)$, Gleason gives the σ -additive measure on the lattice $L(H)$ itself — but this lattice-level rigidity does *not* transfer to the dual: the clustering argument (§2.1) shows the passage to a charge on $S_0(L(H))$ fails for every normal state. Since extension fails there is no dual measure $\hat{\mu}$ to descend, so the descent question does not even arise for normal states on $L(H)$.

Status. Extension axis: settled (finite; infinite $L(H)$ resolved for normal states by Proposition 2.6, the singular case the open sliver). Descent axis: open, three known routes blocked (RDP / MacNeille / σ -Stone), no impossibility theorem. Not a current active lead.

The frontier and the first concrete move. The computational and literature work is exhausted; what remains is human mathematical work, in two branches that share one wall.

(a) The deep unlock

the σ -OML duality prerequisite (descent axis). Build a Loomis–Sikorski-type representation, or a countable-join-preserving σ -Stone duality, for a concrete non-Boolean infinite σ -OML — or prove none exists. All three known routes are blocked but no impossibility theorem rules it out, so the outcome is binary and large: a new representation resolves the cluster, an impossibility result closes it (and both downstream residues) at once. High ceiling, no current foothold.

(b) The tractable special case

the singular-state sliver (§2.1): does a singular orthoadditive state on $L(H)$ extend to a charge on $S_0(L(H))$? A concrete operator-algebra question (Calkin, Bunce–Wright, Takesaki) about one specific lattice, *conditionally* self-contained — see the finite-vs-countable hinge below.

Sequence **(b) first**, for a reason that does not depend on its being easy: it is informative either way. If a singular state fails to extend, then *no* state on $L(H)$ extends, so PR_{dual} fails for all of $L(H)$; since the singular sliver is the only candidate witness for “descent does independent work” (§4, two grades), this closes the PR_{dual} third-grade question negatively — an impossibility-flavoured closure. If a singular state *does* extend, it is the first concrete object that extends but may not concentrate, which hands branch (a) its motivating example. Either outcome advances the cluster.

The hinge that gates (b). Whether (b) is genuinely the easier branch turns on a single fact, *not yet settled*: is the singular-extension obstruction reached with *finite* or *countable* meet-zero families (§2.1)? The normal-state result is finitary and so dodges the descent-axis stability wall; the singular case names “ σ -additivity of the state” as operative, which would make it countable — and then it is not cleanly finitary and does not obviously dodge that wall. Settle finite-vs-countable first: if finite, (b) is a self-contained operator-algebra problem to attack ahead of (a); if countable, (b) is entangled with the same finitary-to- σ bridge as (a) and the tractability advantage evaporates.

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